

|     |                                                                                                                                                                                    |                          |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----|
| 3   | (a) work done bringing unit positive charge<br>from infinity (to the point)                                                                                                        | M1<br>A1                 | [2] |
| (b) | (i) <i>either</i> both potentials are positive / same sign<br>so same sign<br><i>or</i> gradients are positive & negative (so fields in opposite directions)<br>so same sign       | M1<br>A1<br>(M1)<br>(A1) | [2] |
|     | (ii) the individual potentials are summed                                                                                                                                          | B1                       | [1] |
|     | (iii) allow value of $x$ between 10 nm and 13 nm                                                                                                                                   | A1                       | [1] |
|     | (iv) $V = 0.43 \text{ V}$ ( <i>allow</i> $0.42 \text{ V} \rightarrow 0.44 \text{ V}$ )<br>energy $= 2 \times 1.6 \times 10^{-19} \times 0.43$<br>$= 1.4 \times 10^{-19} \text{ J}$ | M1<br>A1<br>A1           | [3] |

|               |                                            |                 |              |
|---------------|--------------------------------------------|-----------------|--------------|
| <b>Page 3</b> | <b>Mark Scheme</b>                         | <b>Syllabus</b> | <b>Paper</b> |
|               | <b>GCE A LEVEL – October/November 2013</b> | <b>9702</b>     | <b>43</b>    |